

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

2 Artificial Intelligence (sbh11)

A *Boolean satisfiability problem* has four variables, x_1 , x_2 , x_3 and x_4 . A literal l can be a variable or its negation, denoted $\bar{l}$. The formula of interest, in conjunctive normal form (CNF), is

$$f = (x_2 \vee \bar{x}_3) \wedge (\bar{x}_2 \vee x_3) \wedge (x_1 \vee x_2 \vee \bar{x}_4). \quad (1)$$

The aim is to find assignments to the variables such that f is true under the usual rules for Boolean operations. This question addresses the use of more general *constraint satisfaction* to solve this problem.

Constraint
satisfaction
problems

- (a) Give a general description of a *constraint satisfaction problem (CSP)*. [3 marks]

Answer: We have a collection of variables, each with a domain. The domain specifies what values a variable can take. We have a set of constraints, each applying to a set of variables and specifying which combinations of values are allowable. We seek an assignment to all the variables, such that no constraints are violated.

Constraint
satisfaction
problems

- (b) Explain how a Boolean satisfiability problem in CNF form and with n variables can be converted to a CSP, also having n variables and having a suitable constraint for each clause. Illustrate your answer using the 4-variable formula f in (1). [3 marks]

Answer: The variables for the CSP are the same as those for the SAT problem. Each still has domain $\{T, F\}$. As all constraints for a CSP have to be upheld, each clause can be turned into a constraint specifying the assignments to its variables that make the clause true. So for example the first clause has a binary constraint on variables x_2, x_3

$$\{(x_2, x_3), (x_2, \bar{x}_3), (\bar{x}_2, \bar{x}_3)\}$$

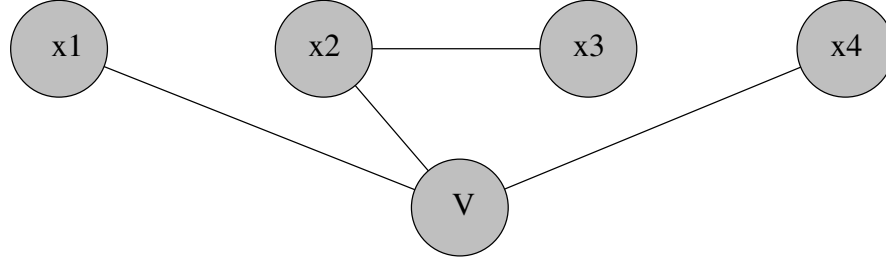
Constraint
satisfaction
problems,
converting to
binary constraints

- (c) Explain, again using a constraint corresponding to a clause from (1), how general constraints can be converted to binary constraints. Provide a graph illustrating the problem from (1) after it has been converted to a CSP with only binary constraints. [4 marks]

Answer: The general method is to introduce new variables to pair with the original variables from the problem. In this case, the only clause that is relevant is the last one. We introduce a new CSP variable V with domain $\{1, 2, 3, 4, 5, 6, 7\}$. The truth table for the third clause is

x_1	x_2	x_4	
F	F	F	T
F	F	T	F
F	T	F	T
F	T	T	T
T	F	F	T
T	F	T	T
T	T	F	T
T	T	T	T

We break the constraint into three binary constraints on (V, x_1) , (V, x_2) and (V, x_4) . Then each line of the truth table that is T adds an acceptable combination to each new constraint. For example, the first row allows combinations $(V = 1, x_1 = F)$, $(V = 1, x_2 = F)$ and $(V = 1, x_4 = F)$, and so on. The graph structure is



Constraint
satisfaction
problems, forward
checking

- (d) Explain, how *forward checking* works in the context of a general CSP. How does this benefit a CSP solver? [3 marks]

Answer: Each time we assign a variable, we examine the constraints it is involved in. If the value assigned makes some value for a connected variable impossible, then we delete that possibility from its domain. This can allow us to detect inconsistencies more quickly because a domain may become empty.

Constraint
satisfaction
problems, forward
checking

- (e) Using the CSP equivalent you developed for (1), with only binary constraints, demonstrate how *forward checking* works using the sequence of assignments $x_1 = F$, $x_2 = F$, $x_4 = T$. [5 marks]

Answer: This has been seen in lectures, but only with graph colouring problems for which the constraints behave differently, and not when a pair of constraints addresses the same pair of variables. The following table summarises the process:

	V	x_1	x_2	x_3	x_4
	$\{1, 2, 3, 4, 5, 6, 7\}$	$\{T, F\}$	$\{T, F\}$	$\{T, F\}$	$\{T, F\}$
$x_1 = F$	$\{1, 2, 3\}$	$= F$	$\{T, F\}$	$\{T, F\}$	$\{T, F\}$
$x_2 = F$	$\{1\}$	$= F$	$= F$	$\{F\}$	$\{T, F\}$
$x_4 = T$	$\{\}$	$= F$	$= F$	$\{F\}$	$= T$

Setting $x_1 = F$ immediately reduces the domain of V , but nothing else is affected. Setting $x_2 = F$ affects constraints with both V and x_3 . Setting $x_4 = T$ removes the last possibility for V and hence implies an inconsistency.

Constraint
satisfaction
problems,
constraint
propagation

- (f) How would you expect the solution obtained when applying forward checking to be affected if constraints were allowed to propagate more widely? [2 marks]

Answer: This is alluding to the idea that by allowing constraints to propagate we might find an inconsistency or solution earlier. In this case the fact that V has domain $\{1\}$ after the second assignment implies that $x_4 = F$. At this point we would have found a solution: all variables F .